

TechMantra Production Server Deployment, Update & Maintenance Documentation

theteachmantra.com | Django + Gunicorn + Nginx + IIS ARR + PostgreSQL

Component	Production Setup
Domain	theteachmantra.com / www.theteachmantra.com
Public Server	103.210.29.175
Linux	Ubuntu 24.04 LTS
Application	Django
Gunicorn	127.0.0.1:8001
Nginx	Linux port 8088
Public Proxy	Windows IIS + ARR
Database	Local PostgreSQL 17
Static	/var/www/techmantra/static
Media	/var/www/techmantra/media

1. Production Architecture

Public HTTP/HTTPS traffic is handled by Windows IIS. IIS ARR reverse-proxies TechMantra over the private LAN to Linux Nginx. Nginx serves static/media directly and forwards application requests to Gunicorn.

```
Internet
-> theteachmantra.com / www.theteachmantra.com
-> Windows IIS / ARR (80/443)
-> Linux 192.168.1.37:8088
-> Nginx
/static/ -> /var/www/techmantra/static/
/media/ -> /var/www/techmantra/media/
/ -> Gunicorn 127.0.0.1:8001
-> Django
-> PostgreSQL 17
```

2. Important Production Paths

```
Project: /home/techmantra/apps/TechMantra
Virtual environment: /home/techmantra/apps/TechMantra/venv
Environment: /home/techmantra/apps/TechMantra/.env.server
Static: /var/www/techmantra/static
Media: /var/www/techmantra/media
Backups: /home/techmantra/backups
Logs: /home/techmantra/logs
Service: techmantra.service
```

Important: Never store production passwords, database credentials, Django secret keys, or other secrets in GitHub or documentation.

3. How to Deploy the Next Website Version

A. Development machine

```
git status
git add .
git commit -m "Describe the website update"
git push origin main
```

B. Connect to production

```
ssh techmantra@103.210.29.175 -p 3000
cd ~/apps/TechMantra
source venv/bin/activate
```

C. Inspect before updating

```
git status
git branch
git fetch origin
git log --oneline HEAD..origin/main
```

Important: Do not blindly run `git pull`. The server contains intentional production-specific safety/configuration changes that must be preserved.

4. Production-Specific Modified Files

```
myapp/context_processors.py
myapp/views.py
myproject/settings.py
myproject/urls.py
myproject/wsgi.py
```

The `.env.server` file, migrated media, production database configuration, static/media paths, and other server-only configuration must also remain protected.

Important: Avoid git reset --hard or git restore on production unless there is a verified backup and the exact impact is understood.

5. Commands After a Code Update

If requirements.txt changed

```
pip install -r requirements.txt
```

If Django models/database schema changed

```
python manage.py migrate
```

Important: Create a database backup before production migrations.

If CSS/JS/static assets changed

```
python manage.py collectstatic --clear --noinput
```

The duplicate static discovery issue has been fixed. Django now discovers myapp/static only once.

Validate application

```
python manage.py check
```

Restart Gunicorn

```
sudo systemctl restart techmantra
sudo systemctl status techmantra --no-pager
```

Use an authorized administrative account or the restricted service-management mechanism when restarting the service.

6. Post-Deployment Verification

```
curl -I https://theteachmantra.com
curl -I https://www.theteachmantra.com
```

Both domains should return a successful HTTP response. Then manually verify homepage layout, key pages, admin/login where applicable, forms, database content, images, CSS/JavaScript, and uploads affected by the release.

7. Static/CSS Troubleshooting

```
cd ~/apps/TechMantra
sha256sum ./myapp/static/css/style.css
sudo sha256sum /var/www/techmantra/static/css/style.css
python manage.py findstatic css/style.css --verbosity 2
```

After collectstatic, source and deployed hashes should match. This check previously identified the stale CSS that caused oversized logos and an incorrect homepage layout.

8. Logs and Diagnostics

```
sudo systemctl status techmantra --no-pager
sudo systemctl status nginx --no-pager
sudo nginx -t

sudo tail -n 100 /var/log/nginx/error.log
sudo tail -n 150 /var/log/nginx/access.log

tail -n 100 /home/techmantra/logs/gunicorn-error.log
tail -n 100 /home/techmantra/logs/gunicorn-access.log
```

9. Backup Rules

Take a fresh backup before risky releases, database migrations, major configuration changes, or storage operations. PostgreSQL application backups should target the public schema.

```
pg_dump -Fc -n public [database options] > ~/backups/techmantra-$(date +%Y%m%d-%H%M%S).dump
```

Important: Do not put database passwords directly in commands that may remain in shell history.

10. Do Not Change These Casually

1. Do not delete/change Vercel until the self-hosted production deployment is intentionally accepted as the replacement.
2. Do not delete Cloudinary resources yet because the old Vercel deployment may still depend on them.
3. Do not expose Linux port 8088 publicly; IIS should access it over the private LAN.
4. Do not grant unrestricted sudo to the techmantra user.
5. Do not commit .env.server to GitHub.
6. Do not replace production PostgreSQL data with the repository SQLite database.
7. Do not run destructive Git commands without a verified rollback point.

11. Recommended Future Deployment Flow

```
Develop -> Test -> Commit -> Push
|
Production: Backup -> Fetch/Review -> Controlled Update
-> Install requirements if changed
-> Migrate if required
-> collectstatic
-> Restart Gunicorn
-> Health check
-> Browser verification
```

A dedicated deployment script can later automate this sequence while preserving production-specific configuration.

12. Current Status

Item	Status
DNS apex	Working
DNS www	Working
IIS / ARR	Working
Nginx	Working
Gunicorn / Django	Working
PostgreSQL production data	Migrated locally
Cloudinary media	Migrated locally
Static files	Local and synchronized
Homepage layout	Verified working
Vercel	Intentionally untouched

Important: Remaining hardening: verify HTTPS proxy headers, CSRF trusted origins, secure proxy handling, SSL renewal/coverage, upload permissions, final backup, and restricted service-management permissions.